Bechir Karmeni

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Proficient in spoken and written English, French, Arabic, and German.

Skills

- **Languages**: Python, C++, C, SQL, CSS, JavaScript, PHP
- **Frameworks**: Scikit-learn, TensorFlow, Keras, Flask, PyTorch, NumPy, Pandas, Matplotlib, PySpark
- **Tools**: Docker, GIT, Power BI, MySQL
- **Platforms**: Linux, AWS

Education

- **Ecole Polytechnique Sousse** Sep 2021 - Jun 2024
  Bachelor's degree in software engineering and Information Technology *Sousse, Tunisia*

  Courses: Machine Learning, Algorithms and data structures, Graphs and optimization, Artificial intelligence, Virtualization and cloud, Design of information systems

Experience

- **ACTIA ENGINEERING SERVICES** Feb 2024 - Jun 2024
  Software Designer (Internship) *Onsite - Tunisia*

  - Engineered a centralized tool for automating and streamlining component obsolescence management, resulting in increased productivity, less manual labor, and errors, and significantly reduced processing time.

  - Created the KPI dashboard with PowerBI to deliver meaningful data visualizations and performance metrics, allowing stakeholders to effectively monitor key performance indicators and make better strategic decisions.

  - Created interactive user interfaces for data visualization with PowerBI integrated into a web platform, significantly enhancing decision-making through real-time data insights.

  Key words : Python, Angular, Sprint Boot, Power BI, Tool, Management of component obsolescence

- **VECTEUR INDUSTRIE** Jan 2022 - Jun 2022
  Front-End Developer (Part-time) *Hybrid - Tunisia*

  - Designed and executed user interface components, fostering a seamless and responsive user experience that boosted overall satisfaction and engagement.

  - Profiled and improved front-end performance, enhancing application speed and efficiency, providing a smoother, more reliable user experience.

  - Documented the front-end codebase, maintaining clarity and ease of future development, facilitating smoother handovers and more efficient ongoing maintenance.

  Key words : User Interface (UI), HTML, CSS, JavaScript

- **ADN EXPERTISE** Jul 2022 - Sep 2022
  Machine Learning Developer (Internship) *Hybrid - Tunisia*

  - Extracted comprehensive data from complex databases, enabling informed, strategic decision-making by providing clear, detailed, and interactive reports that drove business insights and operational improvements.

  - Devised a sophisticated recommender system utilizing advanced machine learning algorithms, markedly enhancing user engagement and satisfaction by providing highly personalized recommendations.

  - Designed the KPI dashboard using PowerBI, which provided insightful data insights and key metrics, facilitating enhanced decision-making and strategic planning across the organization.

  Key words : Power BI, Data Visualization, Data Analysis, Report Extraction, Machine Learning

Projects

MediMap Mar 2023 - Mar 2024
Demo

- Implemented a feature to provide real-time information on nearby mammography facilities, streamlining access and enhancing patient convenience for timely breast cancer screenings.

- Implemented AI algorithms to assess breast cancer risk and personalize screening recommendations, improving accuracy and effectiveness in early detection and prevention.

- Established a remote booking system for mammography appointments, facilitating flexible scheduling and ensuring timely, organized screenings.

- Developed a secure digital medical record platform with controlled access, significantly improving patient data privacy and streamlining record management processes.

MedoPad Sep 2023 - Jan 2024
Report

- Integrated multiple disease detection capabilities into a single interface for streamlined diagnostics.
- Architected and deployed specialized AI models such as CNNs for COVID-19 and pneumonia detection, VGG-16 for feature extraction in brain tumors, Random Forest for breast cancer and diabetes, and XGBoost for heart disease.

Lung Cancer Prediction Apr 2023
presentation

- Utilized UNet, AlexNet, and VGG deep learning models to analyze X-ray images for carcinogenic regions, enabling a broad comparison of their effectiveness.
- Applied lung segmentation and normalization to enhance model accuracy and reliability in detecting carcinogenic regions.
- Employed over 1000 CT scans from the LUNA16 challenge with detailed annotations and metadata, improving model training and testing.
- Reported accuracy, precision, recall, and F1-score, identifying UNet as the most effective model for cancer detection.

Certification

SQL (Advanced) Sep 2024
Issued by HackerRank

Problem Solving (Intermediate) Aug 2022
Issued by HackerRank

Professional Data Analyst Apr 2022
Issued by DataCamp

Awards

First prize win at MutualHack by MAE 2.0 May 2025
Issued by MAE ASSURANCES

Second Place Win at the CivicTech Sprint Hackathon Dec 2024
Issued by The World Bank and Ministry of Interior of Tunisia, in partnership with EY , Deloitte

First prize win at SWAFYTHON Nov 2024
Issued by SWAFY (Science With And For Youth) funded by EU4Youth

Third prize at the third edition of the Tunisia e-Health Valley Hackathon Sep 2024
Issued by OPALIA RECORDATI and Union of Pharmacists of Tunisia)

Second prize win at the OPENGEODATA Challenge Hackathon Nov 2023
Issued by AGEOS (African Association for Geospatial Development)

Third prize win at CYBERBENDERS AI CHALLENGE Nov 2023
Issued by EPI Digital School

Third prize win at the Tunisia e-Health Valley Hackathon Sep 2023
Issued by Opalia Recordati

First prize win at Artificial Intelligence Night Second Edition Apr 2023
Issued by Association for Scientific Research and Innovation in Computer Science

First prize win at Start It Up Hackathon First Edition Mar 2023
Issued by Junior Entreprise Eniso

Silver medal In "Nuit de l'Info" 2022 Dec 2022
Issued by Chartres Métropole Innovations Numériques

First prize win at GIZ Hackathon First Edition Dec 2022
Issued by IEEE Faculty of Scienece Tunis